

# HENRIQUE K. MIYAMOTO

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<https://miyamotohk.github.io/>

## EDUCATION

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| <b>Université Paris-Saclay</b> , France<br>Ph.D. in Information Theory<br><i>Graduate School of Computer Science</i>                                                           | 2022–present |
| <b>Universidade Estadual de Campinas</b> , Brazil<br>M.Sc. in Applied Mathematics<br><i>Dissertation: Geometry, Statistics and Applications to Communications and Learning</i> | 2021–2022    |
| <b>CentraleSupélec</b> , France<br>Diplôme d'Ingénieur (M.Sc.)<br><i>Major: Applied Mathematics</i>                                                                            | 2018–2021    |
| <b>Universidade Estadual de Campinas</b> , Brazil<br>B.Sc. in Electrical Engineering<br><i>Class rank: 1/74, GPA: 0.9196/1.0</i>                                               | 2015–2021    |

## RESEARCH INTERESTS

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Information theory and communications (MSC2020 94A and 94B).

## RESEARCH EXPERIENCE

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2022–present, *doctoral researcher*, L2S, CNRS/CentraleSupélec/Université Paris-Saclay, France.

- Advisor: Prof. Sheng Yang.

2021–2022, *master's researcher*, IMECC, Universidade Estadual de Campinas, Brazil.

- Advisor: Prof. Sueli I. R. Costa.

2020–2020, *researcher intern*, Huawei Technologies, France.

- Supervisors: Dr. Ingmar Land, Dr. Apostolos Destounis and Prof. Jean-Claude Belfiore.

2019–2020, *student researcher*, L2S, CNRS/CentraleSupélec/Université Paris-Saclay, France.

- Supervisor: Prof. Sheng Yang.

2016–2018, *undergraduate student researcher*, IMECC, Universidade Estadual de Campinas, Brazil.

- Supervisors: Prof. Henrique N. Sá Earp and Prof. Sueli I. R. Costa.

## TEACHING TRAINING

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**Université Paris-Saclay**, France  
Certificate “Teaching in Higher Education” (Label “Enseigner dans le Supérieur”) 2022–present

- This program certifies training in different teaching skills, and teaching experience of at least 96h, done in parallel to the preparation of the Ph.D. degree.

## TEACHING EXPERIENCE

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2022–2025, *graduate teaching assistant*, **CentraleSupélec**, France.

- Information theory
- Signal processing
- Communication theory
- Digital signal processing
- Wireless communications

2022–2022, *graduate teaching assistant*, **Universidade Estadual de Campinas**, Brazil.

- Principles of communications I

2021–2021, *undergraduate teaching assistant*, **Universidade Estadual de Campinas**, Brazil.

- Principles of communications I

## PREPRINTS

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- [Z2] H. K. Miyamoto and S. Yang, “Error exponents for randomised list decoding”, *arXiv:2601.09519*, 2026 . doi: 10.48550/arXiv.2601.09519.
- [Z1] H. K. Miyamoto and S. Yang, “Universal decoding over finite-state additive channels via noise guessing”, *arXiv:2501.12971*, 2025 . doi: 10.48550/arXiv.2501.12971.

## JOURNAL PAPERS

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- [J4] H. K. Miyamoto, F. C. C. Meneghetti, J. Pinele and S. I. R. Costa, “On closed-form expressions for the Fisher–Rao distance”, *Information Geometry*, vol. 7, pp. 311–354, Nov. 2024, doi: 10.1007/s41884-024-00143-2.
- [J3] H. K. Miyamoto, F. C. C. Meneghetti and S. I. R. Costa, “The Fisher–Rao loss for learning under label noise”, *Information Geometry*, vol. 6, pp. 107–126, Jun. 2023, doi: 10.1007/s41884-022-00076-8.
- [J2] H. K. Miyamoto and S. Yang, “Context-tree-based lossy compression and its application to CSI representation”, *IEEE Transactions on Communications*, vol. 70, no. 7, pp. 4417–4428, Jul. 2022, doi: 10.1109/TCOMM.2022.3173002.
- [J1] H. K. Miyamoto, S. I. R. Costa and H. N. Sá Earp, “Constructive spherical codes by Hopf foliations”, *IEEE Transactions on Information Theory*, vol. 67, no. 12, pp. 7925–7939, Dec. 2021, doi: 10.1109/TIT.2021.3114094.

## CONFERENCE PAPERS

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- [C9] R. Zhang, H. Miyamoto, R. Combes and S. Yang, “An efficient synchronization scheme for distributed bandits”, accepted to *IEEE International Symposium on Information Theory (ISIT)*, Guangzhou, 2026.
- [C8] H. K. Miyamoto and S. Yang, “On  $\alpha$ -tilted noise guessing decoding”, accepted to *IEEE International Symposium on Information Theory (ISIT)*, Guangzhou, 2026.
- [C7] H. K. Miyamoto and S. Yang, “Error exponents for randomised list decoding”, accepted to *IEEE International Symposium on Information Theory (ISIT)*, Guangzhou, 2026.

- [C6] H. K. Miyamoto and S. Yang, “On universal decoding over discrete additive channels by noise guessing”, *IEEE Information Theory Workshop (ITW)*, pp. 1–6, Sydney, 2025, doi: 10.1109/ITW62417.2025.11240417.
- [C5] H. K. Miyamoto and S. Yang “On universal decoding over memoryless channels with the Krichevsky–Trofimov estimator”, *IEEE International Symposium on Information Theory (ISIT)*, pp. 1498–1503, Athens, 2024, doi: 10.1109/ISIT57864.2024.10619414.
- [C4] F. C. C. Meneghetti, H. K. Miyamoto, S. I. R. Costa and M. H. M. Costa, “Revisiting lattice tiling decomposition and dithered quantisation”, *Geometric Science of Information (GSI)*, Saint-Malo, 2023. In: F. Nielsen and F. Barbaresco (eds.), *GSI 2023*, LNCS 14071, pp. 318–327, Springer, 2023, doi: 10.1007/978-3-031-38271-0\_319.
- [C3] F. C. C. Meneghetti, H. K. Miyamoto and S. I. R. Costa, “Information properties of a random variable decomposition through lattices”, *41st International Conference on Bayesian and Maximum Entropy Methods in Science and Engineering (MaxEnt)*, Paris, 2022. In: *Physical Sciences Forum*, vol. 5, no. 1, 2022, doi: 10.3390/psf2022005019.
- [C2] H. K. Miyamoto and S. Yang, “A CSI compression scheme using context trees”, *International Zurich Seminar on Information and Communication (IZS)*, pp. 24–28, Zurich, 2022, doi: 10.3929/ethz-b-000535273.
- [C1] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, “Constructive spherical codes in  $2^k$  dimensions”, *IEEE International Symposium on Information Theory (ISIT)*, pp. 1612–1616, Paris, 2019, doi: 10.1109/ISIT.2019.8849464.

## DOMESTIC CONFERENCE PAPERS

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- [D3] H. K. Miyamoto, “Geometria, estatística e aplicações a comunicações e aprendizado”<sup>1</sup>, *Proceeding Series of the Brazilian Society of Computational and Applied Mathematics*, vol. 10, no. 1, 2023, doi: 10.5540/03.2023.010.01.0004 (in Portuguese).
- [D2] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, “Construção de códigos esféricos usando a fibração de Hopf”, *Jornada Nacional de Iniciação Científica (JNIC)*, 70ª Reunião Anual da SBPC, Maceió, 2018, ISSN: 2176-1221 (in Portuguese).
- [D1] H. K. Miyamoto, H. N. Sá Earp and S. I. R. Costa, “Construction of spherical codes using the Hopf fibration”, *XXV Congresso de Iniciação Científica da Unicamp*, Campinas, 2017, doi: 10.19146/pibic-2017-78809.

## PATENT APPLICATIONS

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- [P1] I. Land, H. K. Miyamoto, A. Destounis and J.-C. Belfiore, “Transfer Learning between Neural Networks”, WO2022207097 (2022), CN117121020 (2023), EP4309079 (2024).

## THESES

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- [T1] H. K. Miyamoto, “Geometria, Estatística e Aplicações a Comunicações e Aprendizado” (Geometry, Statistics and Applications to Communications and Learning), *M.Sc. dissertation in applied mathematics*, Universidade Estadual de Campinas, Campinas, 2022, hdl: 20.500.12733/6633.

## SELECTED PRESENTATIONS (WITHOUT PROCEEDINGS)

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IEEE European School on Information Theory (ESIT), *poster presentation*, Eindhoven, The Netherlands, 2024.

France PhD Workshop on Information Theory, *poster presentation*, Palaiseau, France, 2024.

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<sup>1</sup>This is an extended abstract of [T1], published on the occasion of the Clóvis Caesar Gonzaga Award (see Awards).

IEEE European School on Information Theory (ESIT), *poster presentation*, Bristol, UK, 2023.

International Conference on Information Geometry for Data Science (IG4DS), *oral presentation*, Hamburg, Germany (virtual), 2022.

**International Congress of Mathematicians (ICM)**, *oral presentation*, Rio de Janeiro, Brazil, 2018.

Latin American Week on Coding and Information (LAWCI), *poster presentation*, Campinas, Brazil, 2018.

Jornada de Matemática, Matemática Aplicada e Educação Matemática (J3M), *oral presentation*, Curitiba, Brazil, 2017.

## AWARDS

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*1st place* in **Clóvis Caesar Gonzaga Award** for master's dissertations in applied and computational mathematics, SBMAC, 2023.

*Merit Award* in Jornada Nacional de Iniciação Científica (JNIC), SBPC, 2018.

*2nd place* in **Beatriz Neves Award** for undergraduate research in applied and computational mathematics, SBMAC, 2018.

*Academic Excellence in Undergraduate Work* in Jornada de Matemática, Matemática Aplicada e Educação Matemática (J3M), category Geometry and Topology, PET Matemática/UFPR, 2017.

*Scientific Merit* in XXV Congresso de Iniciação Científica da Unicamp, PRP/Unicamp, 2017.

*Merit Award* for the composition produced in Unicamp entrance exam, COMVEST/Unicamp, 2015.

## SCHOLARSHIPS

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2021–2022, *Master's Degree Scholarship*, São Paulo Research Foundation (FAPESP), 2021/04516–8.

2018–2020, *Eiffel Excellence Scholarship*, French Ministry of Europe and Foreign Affairs.

2016–2018, *Scientific Initiation Scholarship*, São Paulo Research Foundation (FAPESP), 2016/05126–0.

2013–2013, *Junior Scientific Initiation Scholarship*, Brazilian National Council for Scientific and Technological Development (CNPq).

## ACADEMIC SERVICE

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### Scientific societies membership

2025–present, *member*, Société Mathématique de France (SMF).

2025–present, *member*, Société de Mathématiques Appliquées et Industrielles (SMAI).

2020–present, *student member*, Sociedade Brasileira para o Progresso da Ciência (SBPC).

2018–present, *graduate student member*, Institute of Electrical and Electronics Engineers (IEEE).

- Information Theory Society, Communications Society, Signal Processing Society.

### Reviewing

#### Journals

Information Geometry, Springer

Journal of Communication and Information Systems (JCIS)

#### Conferences

2023–2026, IEEE International Symposium on Information Theory (ISIT)

2023, Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT)

### **Committees and representative positions**

2023–2025, *doctoral and postdoc researchers representative*, Laboratory of Signals and Systems (L2S).

2023, *organising committee*, Encontro de Códigos, Reticulados e Informação, IMECC, Unicamp.

2018–2019, *student representative*, Departamento de Comunicações, FEEC, Unicamp.

### **LANGUAGES**

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| Portuguese | native                   |
| English    | advanced (Cambridge CAE) |
| French     | advanced                 |
| Spanish    | intermediate             |
| Japanese   | elementary (JLPT N4)     |